

Replication of combined school readiness interventions for teachers and parents of head start pre-kindergarteners using remote delivery[☆]

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ABSTRACT

This replication study examined remote delivery of 2 evidence-based prekindergarten (pre-k) interventions to evaluate whether the combination of interventions, one in Head Start classrooms (online The Early Education Model, eTEEM) and one in the home (online Play and Learning Strategies, ePALS) resulted in enhanced effects on at-risk 3- to 5-year-old children's school readiness skills when compared to either of these interventions alone. The aligned interventions trained adults – teachers and parents – to use responsiveness and guided learning approaches within routine activities with the aim of increasing children's social-behavioral and academic skills. Both the original in-person study (Landry et al., 2017) and the present remote study used a four-group factorial design. The present study first randomized classrooms to eTEEM ($n = 36$) or control ($n = 33$). Next, within classrooms, parent-child dyads were randomized to ePALS or No ePALS, resulting in four conditions: $n = 186$ eTEEM/ePALS; $n = 151$ eTEEM/No ePALS; $n = 160$ No eTEEM/ePALS; and $n = 133$ No eTEEM/No ePALS control. The original study found small to moderate effects on parent and teacher behaviors ($d = 0.18$ to 0.55). In this remote study, there were generally larger parent and teacher effects for responsiveness ($g = 0.63$ to 0.70) and for guided learning ($g = 0.37$ to 1.05), behaviors targeted by the interventions. Results of the original study showed few effects on children's academic outcomes and the strongest effect on children's social-behavioral skills for the parent intervention ($d = 0.15$ to 0.19). The present study produced similar patterns of results for child outcomes with larger social-behavioral effects ($g = 0.19$ to 0.27) than academic ($g = -0.23$ to 0.30). We discuss the importance of these replication findings when delivering interventions remotely to improve scalability.

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1. Introduction

This study is a replication of a randomized controlled trial (RCT; Landry et al., 2017) that attempted to reproduce effects on Head Start children's school readiness with a systematic variation of remote delivery for the teacher and parent interventions. The original

study and the present study examined whether training both teachers and parents to use responsive approaches and guided learning resulted in greater social-behavioral and academic skills for children than either intervention alone or a control group. There is recent national interest in replication studies that examine the extent to which intervention effects can be consistently obtained under different conditions (Institute of Education Sciences [IES] and National Science Foundation [NSF], 2018; Kim, 2019). Therefore, the present RCT and the original RCT used equivalent factorial designs with families experiencing poverty and enrolled in Head Start. The present study varied the delivery mechanism of 2 extensively tested interventions - The Early Education Model (TEEM) and Play and Learning Strategies (PALS) - designed for use in pre-kindergarten (pre-k) classrooms and homes, respectively. A barrier in the original study was completion of the parent

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intervention with a 30% attrition rate; we hypothesized that a more convenient, online intervention could improve parent retention.

High-quality experiences in the pre-k classroom and simultaneously at home are critical, as these increase children's long-term social-behavioral (e.g., Rimm-Kaufman & Pianta, 1999) and academic outcomes (e.g., Heckman, Moon, Pinto, Savellyev & Yavitz, 2010). Yet during the pre-k period, the quality of teacher-child and parent-child interactions is often insufficient for children experiencing poverty (e.g., Burchinal, Kainz & Cai, 2011). The federally-funded Head Start pre-k program is designed to raise the quality of both classroom- and home-based experiences for four-year-olds experiencing risk, but effects are inconsistent in magnitude and can fadeout (Burchinal et al., 2011; U.S. Department of Health & Human Services, 2010). Thus, we examined the potential of improving the quality of pre-k school and home experiences using online courses and remote coaching of Head Start teachers, or parents, or both adults.

1.1. Importance of responsiveness, guided learning, and organization during preschool

Both the TEEM and PALS interventions focused on 3 caregiver behaviors that support children's development (see Grusec & Davidson, 2010), including: (a) *responsive* approaches where the adult reciprocally accommodates a child's interests and requests to support positive social behaviors and provide emotional support in stressful situations; (b) *guided learning* that matches task demands to the child's level of understanding via differentiated and scaffolded academic learning; and (c) *organization* that includes predictable routines that give children experiences that support a sense of belonging to a classroom community or their family. These focal adult behaviors are drawn largely from attachment and social learning theories (e.g., Ainsworth, Blehar, Waters & Wall, 1978; socio-cultural: Rogoff, 1990; Vygotsky, 1978). Socially and behaviorally responsive adults provide warm, reciprocal interactions with the child that contingently respond to children's verbal and non-verbal signals of their interests and emotional states (e.g., Dunst & Kassow, 2008). Academically responsive adults respond to children's verbalizations as well children's behaviors during academic tasks to scaffold learning in ways that provide appropriate levels of challenge and support the learning of new concepts and skills (e.g., Tamis-LeMonda, Bornstein & Baumwell, 2001).

Ample research shows that warm, responsive adults can facilitate preschool children's development of *social-behavioral skills* (i.e., prosocial behaviors, self-regulation of emotions, and goal-directed behaviors) in the classroom (e.g., Promoting Alternative Thinking Strategies [PATHS]; Domitrovich, Cortes & Greenberg, 2007) and home environment (e.g., Incredible Years Preschool Program; Presnall, Webster-Stratton & Constantino, 2014). There is also a strong research base that shows adults can guide learning to promote *academic skills* during pre-k that include language, literacy, and math skills. School readiness interventions in pre-k classrooms that improve children's academic skills often use targeted curriculum, student progress monitoring measures, strong models of professional development (PD), and coaching (e.g., Bierman et al., 2008; Lonigan et al., 2015; Weiland & Yoshikawa, 2013). Academic interventions in the home that encourage parents to talk, read, and count with their children can improve pre-k academic skills (e.g., Justice & Ezell, 2000; Niklas, Cohrsen & Tayler, 2016; Sheridan, Knoche, Kupzyk, Edwards & Marvin, 2011). Findings from a number of studies also show the benefit of adults providing *organizational supports* for young children through predictable schedules and routines. Examinations of the effect of small-group instructional routines on academic skills highlight gains in pre-k children's literacy and lan-

guage skills (Farver, Lonigan & Eppe, 2009; Landry, Assel, Carlo, Williams & Wu, 2019) and social and behavioral skills, including decreased school avoidance (Landry et al., 2019). It is hypothesized that the structure and predictability of the small group rotations support the development of academic skills and sense of belonging. Involvement in routines (e.g., dinners, storytelling, singing) in the home also benefits pre-k children, with prior research demonstrating that more frequent participation in family routines contributed to higher ratings of social-emotional skills (Muñiz, Silver & Stein, 2014).

The present study combines support for social-behavioral and academic skills, as a number of studies demonstrate that these areas can be targeted simultaneously to support children's school readiness. For example, a study of a comprehensive preschool curriculum that emphasized organization and routines showed effects on children's social-behavioral skills ($d = 0.32$) and academic skills – language ($d = 0.19$), literacy ($d = 0.30$), and math ($d = 0.25$; Lonigan et al., 2015). Thus, the present study supported multiple domains of development and experimentally contrasted delivering these supports by parents and teachers.

1.2. Rationale for aligned teachers and parent interventions

Coordinating research-based approaches across school and home may enhance the impact of interventions because children can have similar experiences that promote language, academic skills, and behavior regulation across 2 key microsystems where development occurs (Bronfenbrenner & Morris, 2006). Alignment of programs across the home and school contexts could produce synergistic benefits because children receive contingent responses integrated with rich language and academic input across the 2 environments where they spend the greatest amount of time (Bierman, Welsh, Heinrichs, Nix & Mathis, 2015; Landry et al., 2017). Thus, the increased intensity and consistency from aligned programs in the home and classroom might be expected to support children's learning to a greater extent than either program alone.

There are few studies that have implemented a school readiness intervention in both the classroom and the home. In one study, Bierman and colleagues examined the impacts of coordinated Head Start interventions across the school and home. Using the Research-Based, Developmentally Informed-Parent (REDI-P) program and REDI Classroom (REDI-C) program, they worked with 200 parents, children, and teachers in 24 Head Start centers to provide aligned classroom and home interventions contrasted with a classroom only condition. The classroom intervention taught language and literacy skills and used the PATHS social-emotional curriculum (Domitrovich et al., 2007). Immediate child impacts for REDI-C (Bierman et al., 2008) were small for child outcomes ($d = 0.15$ vocabulary; $d = 0.16 - 0.29$ literacy; $d = 0.21 - 0.28$ social-behavioral). The parent approach included 10 home visits in pre-k and 6 in kindergarten and helped parents learn how to support academic and social skills and encourage routines that support learning (e.g., school attendance, sufficient sleep). Impacts in kindergarten showed no group differences in home observations and parent-child tasks, but showed small increases in parent-reported reading and conversation frequency ($d = 0.27 - 0.28$; Bierman et al., 2015). Follow-up studies showed sustained effects at Grade 3 for children who received the REDI-P in academic ($d = 0.28 - 0.29$) and social-behavioral skills ($d = 0.35$). This and other empirical evidence from shared book reading interventions (e.g., Lonigan & Whitehurst, 1998; Roberts, 2008) and mathematics programs (e.g., Clements & Sarama, 2008; Starkey, Klein & Wakeley, 2004) suggest that early, aligned intervention delivered by teachers and families of low-income students can increase academic outcomes.

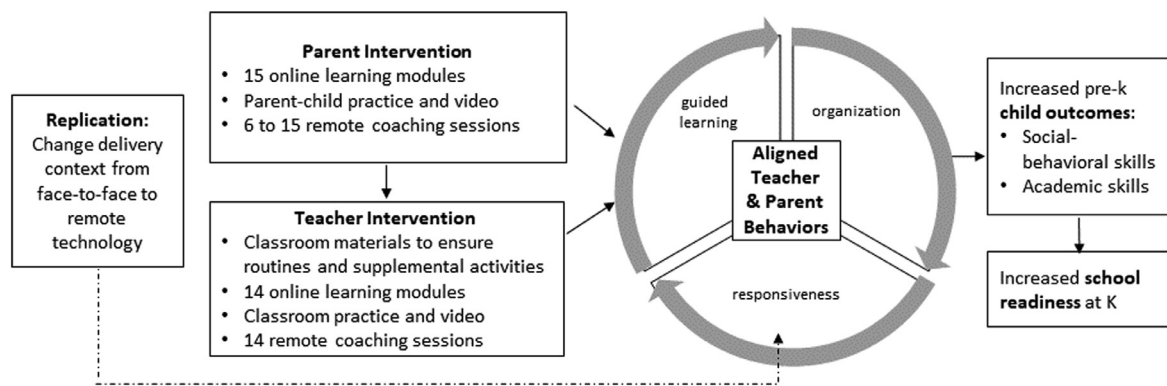


Fig. 1. Intervention Theory of Change for Aligned Teacher and Parent Interventions.

1.3. Practical benefits of remote intervention delivery

Although a small number of studies examine the effects of coordinated teacher and parent interventions, few have undertaken remote delivery approaches to examine if this improves accessibility of evidence-based approaches. Benefits of remote delivery models include increased convenience, scalability, and individualization (Comer et al., 2017; Taylor et al., 2005). Online PD can also increase participant attendance and/or engagement with resources (Fishman et al., 2013; Snell, Hindman & Wasik, 2019). Online PD programs may be more scalable compared to in-person PD programs that are scheduled during the typical work/school day and require transportation between sites (Dede, Ketelhut, Whitehouse, Breit & McCloskey, 2009). This scalability is possible, in part, because remote delivery offers opportunities for immediate individualized resources and/or support that can be accessed in a variety of formats and at the learners' convenience (Bates, McClure, Ross & Womack, 2019).

Virtual coaching, as facilitated through remote delivery approaches, can also allow parents and teachers to review and reflect on video recordings of their interactions with children (Balldin et al., 2016). Video reflections can help parents and teachers attend to specific nuances in interactions that may have otherwise been difficult to remember or observe (Stover, Kissel, Wood & Putman, 2015) and allow for increased levels of active participation and engagement (Dede et al., 2009). For example, findings of one study suggest that teachers who were typically quiet in face-to-face settings were more participatory in technology-mediated interactions (Dede et al., 2009).

1.4. Intervention framework & rationale for replication

The theory of change for both the classroom and home interventions sought to improve adult practices that support children's social-behavioral and academic skills. Social-behavioral approaches focused on positive engagement in social learning activities and self-regulation skills. Academic foci included activities to support children's language, literacy, and mathematics skills. A compelling intervention feature is the combined supports for children's academic school readiness and social-behavioral readiness for learning. As illustrated in Fig. 1, we expected the intervention to impact children indirectly, via improvements in teacher and parent practices.

1.4.1. Past pals studies (face-to-face delivery)

Past PALS studies (Landry et al., 2011; Landry, Smith, Swank & Guttentag, 2008), have separately demonstrated improvements in parents' responsiveness and guided learning attributed to 3 key components: (a) educational modules with videos of other parents

demonstrating the target strategies to increase parents' knowledge and skills, (b) video-taped, in-home practice of the strategies, and (c) coaching and self-reflection of the strategies with opportunities for goal setting and coaching. Parents who participated in PALS were more effective in providing guided learning with rich language input and scaffolded explanations that, in turn, increased children's vocabulary and receptive language skills (e.g., Landry et al., 2008).

1.4.2. Past TEEM studies (face-to-face delivery)

In past TEEM studies (Landry et al., 2011, Landry and colleagues have demonstrated teaching practices improve with 4 key components: (a) provision of classroom learning activities and resources, (b) coursework to increase pedagogical knowledge and skills, (c) classroom practice with opportunities for goal setting and coaching, and (d) training to use child screening and progress monitoring measures to tailor learning to children's individual levels. Teachers who participated in TEEM improved their language and literacy instruction; children supported by TEEM trained teachers showed greater gains in language and literacy skills (Landry et al., 2008, 2011).

1.4.3. Past ePALS studies

We have successfully delivered the remote PALS program (ePALS) in past studies with younger children than the current study. To increase access to early parenting intervention for mothers with infants, we designed an online version of the Infant ePALS program, utilizing remote coaching via phone, based on weekly uploaded parent-infant interaction videos (Feil et al., 2008). Our remote delivery efforts focused on maintaining the effective ingredients of the original PALS intervention and making adaptations to promote ease of use and effectiveness. We maintained the original PALS session content, the presence of a supportive and consistent parent-coach relationship, video modeling of skills within session instruction, video-recorded practice sessions of mothers practicing target skills with her infant for weekly reflective coaching, and weekly skills practice homework. Two randomized control trials of our Infant ePALS program with low income mothers resulted in significant group differences in favor of ePALS with improvements for maternal responsiveness and language support behaviors and infant language and engagement (Baggett et al., 2010; Feil et al., 2020). Results on engagement, feasibility, and consumer satisfaction ratings were very high. Results suggested that the Infant ePALS program was effective as a remotely delivered intervention for economically disadvantaged families and strengthened early parenting behaviors that promote infant social communication and child language development.

1.4.4. Past eTEEM studies

We have also extensively delivered the broader remote TEEM program (eTEEM) through a partnership between the researchers and the local State Education Agency (SEA; Crawford, Zucker, Van Horne & Landry, 2017). In order to increase access to remote coaching for the eTEEM model, we first developed and validated online coaching tools (Crawford, Zucker, Williams, Bhavsar & Landry, 2013). Then, we focused on maintaining the effective ingredients of the original TEEM intervention while promoting ease of use with the SEA partnership. We maintained the TEEM online course content focus and also broadened our online course catalog to include additional topics. All online courses contain extensive video models of effective instructional practices enacted in early childhood classrooms. We also developed tools to ensure supportive, trusting teacher-coach relationships via video conference calls that included time for co-viewing and reflecting on videotaped classroom activities. In addition, all courses included a requirement of weekly skills practice homework that focused on using new behavioral approaches and/or instructional planning. A recent randomized control trial of the eTEEM model (Crawford et al., 2020) contrasted teacher and child outcomes for remote coaching, face-to-face coaching, and business-as-usual control. Both intervention groups received equivalent support for teachers (online child progress monitoring tools, and curricular resources), but the coaching delivery approach was varied. At posttest, regardless of coaching modality, intervention teachers were rated higher on broad classroom instructional quality ($g = 1.09$ on Teacher Behavior Rating Scale) and on language and literacy instructional quality ($g = 0.94$, Early Language and Literacy Classroom Observation Tool). eTEEM teachers' growth in instructional quality related to indirect gains on children's literacy and language skills, but there were no impacts on children's social-behavioral outcomes.

1.4.5. Remote replication aims

Multiple types of replication studies and terminology exist (e.g., Coyne, Cook & Therrien, 2016; Hedges & Schauer, 2018; Schmidt, 2009). We use the terminology *conceptual replication* (see IES & NSF, 2018) for this study that sought to determine if similar effects were found when we systematically varied certain aspects of an earlier efficacy study (Landry et al., 2017). This approach differs from a direct replication study in which the exact same methods and procedures are used across 2 samples. To better understand conditions in which these interventions may be effective, we systematically varied intervention implementation delivery using online learning and remote coaching via technologies, rather than face-to-face courses and coaching approaches used in the original study. We held constant as many factors as possible across the original and present study, including: (a) recruitment in a similar Head Start population that was randomized in the same manner to 4 conditions with a factorial design; (b) implementation under controlled conditions in which the intervention developers provided the training and coaching, rather than other researchers or under less controlled conditions; and (c) a parallel set of outcome measures across studies, with added measures in the area of social-behavioral skills that showed most promise in the earlier study.

1.5. Present study goals & research questions

The aim of the current study was to evaluate the impact of parent and teacher interventions on preschoolers' social-behavioral and academic learning. A randomized control design where intervention conditions were completely crossed, as in the original study (Landry et al., 2017) was used to replicate the original face-to-face randomized control trial in a technology-based, remote delivery format. Our specific research questions (RQ) follow.

1.5.1. RQ1: adult impacts

First, what were the impacts of the eTEEM intervention on teachers' target behaviors? Second, what were the impacts of ePALS on parents' behaviors? We hypothesized the eTEEM intervention would result in greater quality and quantity of targeted teacher responsiveness behaviors and academic instruction during classroom observations. We expected the ePALS intervention would increase targeted parent behaviors of warm, responsiveness and learning support during shared book reading and toy play activities.

1.5.2. RQ2: child outcomes

To what extent did each intervention and combined interventions increase children's school readiness skills including social-behavioral skills (self-regulation, emotion understanding, social competence) and academic skills (language, literacy, and math)? As part of this question we looked for possible synergistic effects (i.e., statistical interactions) when intervening in home and school. Relative to the control group, we expected greater posttest controlling for pretest scores on child outcomes when children's caregivers were trained in eTEEM, ePALS or both interventions. We also expected positive statistical interactions that could suggest a synergism when intervening across the 2 responsive settings.

2. Method

2.1. Research design & participants

This study used a 4-group 2×2 factorial design, contrasting the presence/absence of parent and teacher interventions and combined interventions with a control group. This design replicates the original randomized control trial (Landry et al., 2017) and varies only the intervention delivery. We recruited 40 Head Start schools in an urban area in the Southeast United States across 3 academic years. We recruited one classroom per school ($n = 69$). First, 36 classrooms were randomized to eTEEM and 33 to No eTEEM/Control. Second, consented parent-child dyads in all classrooms were randomly selected and assigned to ePALS or No ePALS. Of the initial 832 total children and parents with consent, 4 to 7 children per teacher were selected and assigned to ePALS and 2 to 6 children per teacher assigned to No ePALS conditions (7–11 total children per classroom). The final parent-child sample ($n = 630$) included 4 conditions at the parent-child level: No eTEEM and No ePALS (parent-child: $n = 133$); No eTEEM but ePALS (parent-child: $n = 160$); eTEEM but No ePALS (parent-child: $n = 151$); eTEEM and ePALS (parent-child: $n = 186$). Teachers were blind to parent condition.

2.1.1. Teachers

Teacher demographics ($n = 76$) are in Table 1 (there are more than 69 initial teachers because of teacher changes). All teachers, except one, were females and most reported African American (57.8%) or Latino/Hispanic (28.2%) race/ethnicities. Almost all teachers had college experience, reporting at least an Associates degree or higher (94.2%); half of teachers had a Bachelor's degree (52.8%). On average, teachers reported 11.4 years ($SD = 6.9$) of teaching experience. Most teachers spoke English in the classroom (74%), or both English and Spanish (26%). There were no significant differences in distributions across eTEEM/control conditions for teacher: race/ethnicity (chi-square (5) = 3.30, $P > 0.05$), education distributions (chi-square(5) = 10.88, $P > 0.05$), or years of teaching experience ($F(1, 62) = 0.16$, $P > 0.05$).

2.1.2. Parents

Parent demographics ($n = 630$) are reported in Table 1. Mothers were the main caregiver for 93% of the children (fathers 5%,

Table 1

Teacher/classroom, parent, and child characteristics by group.

Variable	Teacher/classroom			Parents			Child				
	eTEEM	no eTEEM	Total	ePALS	No ePALS	Total	eTEEM/ePALS	eTEEM/no ePALS	ePALS/no eTEEM	Control	Total
Age in years <i>M</i> (<i>SD</i>)				36.83 (10.78)	37.56 (11.17)	37.17 (10.96)	4.17 (0.56)	4.25 (0.51)	4.26 (0.48)	4.14 (0.52)	4.20 (0.52)
Female (%)	97.1	100.0	98.6	94.8	94.6	94.7	47.9	53.0	48.8	47.4	49.5
Race/Ethnicity (%)											
African American	55.9	63.3	57.8	34.2	35.9	35.0	30.3	31.4	37.3	45.0	35.5
Hispanic/Latino	29.4	20	28.2	52.9	56.5	54.9	62.4	62.9	51.6	45.7	56.2
Caucasian/White	8.8	6.7	7	2.2	3.0	2.6	1.1	2.1	0.7	1.6	1.3
Asian	2.9	6.7	4.2	0.9	0.7	0.8	0.6	0.0	0.0	1.6	0.5
Multi-Racial	2.9	0.0	1.4	8.1	5.6	6.7	5.6	3.6	10.5	6.2	6.5
Other	0.0	3.3	1.4								
Years of teaching <i>M</i> (<i>SD</i>)	11.20 (7.58)	10.64 (6.29)	10.95 (7.02)								
Highest level of Education%											
< High School Diploma				23.4	26.3	24.7					
High School Diploma/ GED	0.0	6.7	2.9	29.4	25.9	27.8					
CDA Credential	0.0	3.3	2.9	–	–	–					
Some College				27.5	29.5	28.4					
Associates Degree	18.2	13.3	14.3	8.6	8.0	8.3					
Bachelor's Degree	67.7	40.0	52.8	8.6	7.1	7.9					
Some Graduate School	9.1	6.7	10.0	1.5	0.5	1.0					
Master's degree or higher	6.1	30.0	17.1	1.1	2.7	1.8					
Classroom/home language (%)											
Mostly English	73	75	74				44.6	51.7	50.6	48.9	48.7
Both English and Spanish	27	25	26				23.7	20.5	21.9	19.6	21.6
Mostly Spanish							26.3	25.8	21.3	24.8	24.6
Other							5.4	2.0	6.3	6.8	5.1
Free and reduced lunch (%)							95.7	94.9	96.9	96.3	95.9

Note. eTEEM, The Early Education Model; ePALS, Play and Learning Strategies; Control group = No TEEM/No PALS group.

others including grandparents 2%). The most common reported race/ethnicity was White-Hispanic (54%). Nearly a quarter (24%) of main caregivers reported no high school diploma or GED, and only 11% held a Bachelor's degree. Median age was 31.1 years (range 18.7 to 72.3 years), and 64% of main caregivers reported that they contribute to household income. There were no significant differences in parent demographic variables across conditions.

2.1.3. Child

Child ($n=630$) demographics are also reported in Table 1. Children's mean age on September 1st (of the year they participated) was 4.20 years ($SD=0.52$, Range: 2.93 - 5.67), and 50% were female. Many children spoke English as their primary home language (48.7%). The most common reported race/ethnicity was White-Hispanic (56.2%). There was one statistically significant difference in demographics by assignment: a greater proportion of children in the eTEEM conditions were White-Hispanic than those assigned to No eTEEM conditions, chi-square (12) = 22.18, $P < 0.05$. Online Appendix Tables A1-A2 detail the present sample and original study demographics; these samples were similar across studies.

2.1.4. Attrition

At the teacher/classroom level, eight teachers (11%) attrited across the course of the study (4 eTEEM and 4 control). After one teacher withdrew, this teacher/classroom was never replaced; 7 teachers/classrooms were replaced such that replacement classrooms were assigned to the same condition as the dropped teacher's classroom. Thus, across the study 76 teachers participated in 76 unique classrooms, but after these changes there were 68 classrooms included at posttest (36 eTEEM/32 No eTEEM). The 10.5% of teachers who attrited differed from stayers on education level ($P < 0.05$, Fisher's exact test), with attrited teachers collectively having higher education levels (e.g., attrited teachers all held a Bachelor's degree or higher) relative to teachers who stayed in the study. No other demographics were significant.

At the parent-child level, 178 parent-child dyads attrited (28%), leaving 452 of the original 630 families. Attrition was similar across conditions (chi-square = 8.82, $P = 0.45$). Final participants at posttest were: $n=102$ control (No eTEEM No ePALS), $n=117$ No eTEEM/ePALS, $n=112$ eTEEM/No ePALS, and $n=121$ eTEEM/ePALS. Analyses of pretest differences between those who attrited and stayed were not significantly different across demographics (e.g., parent education, ethnicity, home language; P 's > 0.05 Fisher's Exact test).

2.2. Measures

As in the 2017 Landry et al. study, we assessed multiple domains and used the same standardized child test battery including *academic skills* - literacy, language, and math - and *social-behavioral skills* - self-regulation, social and emotional functioning. Parent and teacher behaviors were assessed via observation. Assessors were trained by senior research staff across multiple training days for each cohort and required certification before test administration. Classroom instruction was mostly in English; therefore, literacy measures were completed in English. However, if parents indicated that children spoke and understood Spanish at home ($n=276$), children received language measures in both English and Spanish. Self-regulation and math measures were administered in the child's preferred language. Preferred language was determined following an examination of multiple factors (e.g., child performance on English versus Spanish language measures, teacher reports, and observation of children's preferred language during social conversation). There were no significant differences on pretest measures across groups, and small effect sizes (Median = 0.03).

2.2.1. Literacy

We administered 2 subtests of the *Test of Preschool Early Literacy* (TOPEL; Lonigan, Wagner, Torgensen & Rashotte, 2007): Print Knowledge and Phonological Awareness. The Print Knowledge subtest (36 items) measures written language conventions, letter names, and letter sounds. The Phonological Awareness subtest (27 items) measures early blending and elision skills. Internal consistency of the measure is strong with average coefficient alphas across 3–5 year old children of 0.95 and 0.87 these subtests (Lonigan et al., 2007).

2.2.2. Language

We used 2 language measures available in English and Spanish. The *Expressive One-Word Picture Vocabulary Test* (EOWPVT; Brownell, 2000; EOWPVT-Spanish Bilingual Edition, Brownell, 2001), is a norm-referenced test of expressive vocabulary. Children are asked to label pictures of objects, actions, or concepts/categories. Internal consistency values for 2- to 5-year-olds range from 0.93 to 0.95 for Cronbach's alpha values. The *Preschool Language Scale-4th Ed.* (PLS-IV; I. L. Zimmerman, Steiner & Pond, 2002; PLS-IV Spanish, 2002) assessed complex receptive and expressive language. Internal consistency ranges from 0.91 to 0.93 (I. L. Zimmerman et al., 2002).

2.2.3. Math skills

We used the *Test of Early Mathematics Ability-Third Edition* (TEMA-3; Ginsburg & Baroody, 2003) to evaluate early math skills (numeracy, operations, etc.). TEMA-3 is a norm-referenced English test; we used a modified Spanish version. The test publishers report good internal consistency for 4- to 6-year-olds (0.93 to 0.95) and validity.

2.2.4. Self-regulation skills

We administered 4 self-regulation measures. In the *Gift Delay-Wrap task* (e.g., Kochanska, Murray & Harlan, 2000), children were told that they would receive a present but they could not turn and look/peek while a research associate was noisily wrapping the present for 60 s. Next, in the *Waiting for Bow* task, this present was placed in front of the child and they were told not to touch the gift until the examiner returned with a bow with a 2 min delay/wait time. These tasks were videotaped and scored for latency to peek in the Gift Delay-Wrap task and latency to touch in the Waiting for Bow task. These latency scores are highly correlated with other measures of self-regulation (e.g., Merz et al., 2014; Montroy et al., 2019). Inter-rater reliability was high in this sample (0.93, 0.99 at pretest for latency to peek and touch, respectively; both score 0.96 at posttest). These measures are considered “hot” self-regulation measures, capturing a child's ability to inhibit, or delay a response in the presence of a highly desirable reward (e.g., Montroy et al., 2019).

Children completed 2 “cool” self-regulation measures. First, in the *Bear/Dragon task* (Carlson, 2005; Garon, Bryson & Smith, 2008), children were told to follow the “nice” bear puppet's commands (e.g., touch your nose) but not the “mean” dragon puppet's commands. After practice trials with feedback, there are 12 test trials (2 sets of 6 trials, with a reminder of the rules in the middle). The bear and dragon commands alternate in pseudo-random order. Children's responses were coded with scores of 0 = no movement, 1 = partial movement, or 2 = full movement for the bear and dragon separately. Children's performance on “dragon trials” served as the outcome measure. Interrater reliability (ICC = 0.99 at pretest and posttest) and internal consistency were high (Dragon trial Cronbach's alpha 0.97 pretest and 0.96 posttest). Second, in the *Dimension Change Card Sort* (DCCS) task (Frye, Zelazo & Palfai, 1995; Zelazo et al., 2003), children are shown 2 boxes (one blue boat label, second red rabbit). Children sort a set of cards with the

same pictures as the boxes, first sorting by shape (i.e., rabbit or boat). After correctly sorting 5 consecutive cards by shape, rules switch to sort by color (i.e., red, blue). We reminded children of the sorting rule before each trial. Among the 5 “post-switch” trials that were the outcome variable, 2 trials were compatible with the initial shape rule/trial, but 3 were incompatible/color rule. Internal consistency was 0.73 in this sample.

2.2.5. Child social and emotional functioning

Children completed a direct assessment of emotion knowledge and caregivers completed 3 questionnaires. The *Emotion Understanding* measure was designed by Izard (1971) and procedures were adapted by Russell and Bullock (1985) for young children by focusing on core emotions (e.g., angry, sad, happy, scared). Children’s ability to label emotions (expressive) and identify emotions (receptive) depicted in pictures was assessed using eight pictures (e.g., “How does this child feel?”). The second portion required situational emotion understanding. Examiners read a short, emotional vignette and asked the child to label one of 4 pictures that best matched the story (e.g., “Show me a picture of the child who just got a nice, new toy; just what they wanted”). In the current study, internal consistency was good for the expressive task ($\alpha = 0.82$ and 0.78 at pretest and posttest) and receptive ($\alpha = 0.72$ and 0.76), but relatively low for situation task ($\alpha = 0.62$ and 0.55).

Both caregivers (i.e., parents and teachers) completed the *Children’s Behavior Questionnaire* (CBQ; Rothbart, Ahadi, Hershey & Fisher, 2001) Attention Focusing and Inhibitory Control scales. Items are rated on a Likert scale ranging from 1 (extremely untrue of the child) to 7 (extremely true of the child). After reverse scoring select items, these 2 scales are combined to an effortful control composite. The authors report adequate to strong internal consistency ranging from 0.64 to 0.92 (average = 0.77) and interrater reliability from 0.67 to 0.92. Parents and teachers also completed the *Social Competence and Behavior Evaluation* (SCBE; LaFreniere & Dumas, 1996) that contains 3 subscales with 10 items each: Social Competence, Anger/Aggression, and Anxiety/Withdrawal. Items are rated on a Likert scale from 1 (never) to 6 (always). Cronbach’s alphas for the total score in this study were 0.83 at posttest.

Teachers completed the *School Liking and Avoidance Questionnaire* (SLAQ; Ladd & Dinella, 2009), which evaluates children’s school engagement. The SLAQ utilizes a Likert scale ranging from 1 (almost never) to 5 (almost always). The 13-item survey has 2 subscales: school avoidance (6 items; e.g., complains about school) and school liking (7 items; e.g., enjoys most classroom activities). The authors report moderate to high internal consistency; Cronbach’s alphas for this study were greater than 0.78 at posttest.

2.2.6. Teacher behaviors

There were 3 classroom observation measures and one teacher report questionnaire. First, the *Teacher Behavior Rating Scale* (TBRS; Landry, Crawford, Gunnewig & Swank, 2001) assessed changes in target teacher behaviors. TBRS scores represent both quantity and quality of instruction and cover the following content areas (a) sensitivity/responsive behaviors; (b) lesson plans/dynamic assessments; (c) centers; (d) book reading behaviors; (e) print and alphabet knowledge; (f) phonological awareness; (g) written expression; (h) oral language, and (i) team teaching. High inter-rater reliability was documented for this sample (range = 0.80 to 0.98) and internal consistency exceeds 0.70 for all scales.

Second, the quality of teachers’ language and literacy environment and activities was also assessed with 2 subtests from the *Early Language and Literacy Classroom Observation* (ELLCO; Smith, Dickinson, Sangeorge & Anastasopoulos, 2002): (a) General Classroom Environment (GCE); and (b) Language, Literacy, and Curriculum (LLC). The GCE is based on ratings of classroom materials, organization, and management. The LLC is based on ratings of

oral language facilitation, approaches to book reading, and other aspects of literacy instruction. In the current study, the ICCs for the GCE and LLC were 0.80 and 0.62, respectively. The total score summed all 19 items on the 2 subtests.

Third, the *Classroom Environment Checklist* (CEC) is an observational tool comprised of 21 items that focus on the presence and quality of literacy and print centers (including materials), instructional planning materials, and the overall classroom management system. Classroom observers were trained in a multi-day training with senior research staff using a series of classroom videos to practice reliability on the 3 classroom observation measures. Observers were reliable and eligible to collect study data when percent agreement scores were $\geq 80\%$ with a gold standard coder (i.e., senior research coordinator in charge of training).

2.2.7. Parent-child behavioral outcomes

The quality of the parent behaviors was evaluated during 2 video recorded parent-child interactions – *Book Read and Free Play*. Coding of the video recorded interactions was supervised by the study’s principal investigator (PI). Raters were blind to condition (i.e., intervention status). Coders spent approximately 3 weeks in training and were not considered to be reliable until they had achieved a percent agreement score of $\geq 80\%$ with the PI. Ongoing training meetings and reliability checks occurred throughout the project (e.g., half day meetings) to minimize coder drift. These scales have been used extensively in previous research and found to have high reliability and validity (e.g., Landry et al., 2008). Caregiver and child total scores in this sample have good internal consistency (Caregiver: $\alpha = 0.75$, 0.71; Child: $\alpha = 0.84$, 0.76 at pretest, posttest).

Ratings were made on a 5- point scale with higher values indicating better parenting quality (e.g., 1 = almost never; 5 = almost always). During the Book Read task, caregivers were rated on 5 behaviors: language building strategies, book comprehension techniques, enthusiasm and engagement, contingent responsiveness, and intrusiveness/tone. Children were rated on 2 behaviors: book reading engagement and language use. Caregiver-child pairs were rated together on shared enjoyment. The quality of parental interactions during Free Play was coded and included contingent responsiveness, verbal content-label frequency, verbal content-label quality, verbal scaffolding, and warmth. Parental negativity (e.g., intrusiveness) was also coded but had very little variation (i.e., SD close to 0) and was not analyzed. Child behaviors coded during the Free Play setting included: language use, goal directed behavior, responsiveness, and social initiative and communication.

2.2.8. Environment routines

Teachers and parents completed an adapted *Confusion, Hub-bub, and Order Scale* (CHAOS; Matheny, Wachs, Ludwig, & Phillips, 1995) to measure perceptions of the level of chaos in their classroom/home. Respondents rate 16 statements about their environment (e.g., “It’s a real “zoo” in our classroom/home.”) on a 5-point Likert scale (*Strongly Disagree* to *Strongly Agree*). Six items have positive wording (e.g., “There is very little commotion in our classroom/home.”) and are reverse scored so higher total scores indicate more chaos. Internal consistency was good (e.g., classroom $\alpha = 0.79$ and 0.82 at pretest and posttest).

2.3. Classroom intervention procedures

Teachers assigned to eTEEM participated in multiple activities: (a) attending a 2-day kickoff, face-to-face training; (b) administering online progress monitoring measures to students at the beginning-, middle-, and end-of-year; (c) using supplemental instructional resources in their classroom; (d) participating in 14 online course sessions; and (e) engaging in up to 14 individualized

remote coaching sessions. At the kickoff training, teachers were loaned tablet devices (iPad with stand), practiced accessing online resources, learned to use instructional kits/activities, and practiced administering student measures. Teachers received PD certificates and incentive pay for completion of all training and assessment activities.

Throughout the academic year teachers were supported to use the intervention resources. To provide personalized support and targeted small-group instruction to students scoring below benchmarks, teachers were trained to use the CIRCLE (Center for Improving the Readiness of Children for Learning and Education) Progress Monitoring system (CPM; S. H. Landry et al., 2014). CPM included several online subtests (rapid letter naming, vocabulary naming, phonological awareness, math). All schools already had a core curriculum (e.g., Creative Curriculum, Frog Street Press), so only supplemental resources were provided including: (a) social-behavioral posters/manipulatives to follow a routine daily schedule and manage classroom activities (HATCH “Positive Beginnings” kit); (b) academic supports including manipulatives and center activities (Lakeshore “School Readiness Kit”); (c) over 100 supplemental activities (CIRCLE Activity Manual; CIRCLE, 2010) across all domains (i.e., social-emotional, literacy, language, math, science); and (d) interactive whole-group, read-aloud activities (Developing Talkers version 1; Zucker, Solari, Landry & Swank, 2013).

Teachers were also invited to PD opportunities that included 14 online course sessions (i.e., eCIRCLE) and 14 coaching sessions, as detailed in schedule in Online Supplementary Appendix Table A3. Courses rely heavily on expert commentary, videos of authentic high quality teaching practices, and embedded activities/assignments that help teachers apply course concepts in their own classrooms. The proportion of course time was 43% literacy, 29% language, 14% general pedagogy, 7% math, and 7% social-behavioral. Coaching focused on the course topics and occurred approximately biweekly from November to April, requiring about 90 minute per cycle or 3 hour per month. Each coaching cycle used these steps: (a) review an uploaded video of classroom interactions, (b) reflect on teacher behaviors and student responses, (c) select areas for improvement and set goals, and (d) plan action items to achieve goals. The coach edited the video to annotate selected clips with text boxes or voiceover feedback on effective approaches and missed opportunities. This edited video was returned to the teacher for written reflection before the coaching call (30 minute). During a video conference call, the teacher and coach would co-view the edited video, discuss observations, and set goals of behaviors the teacher wanted to improve (60 minute). The coach used a checklist of evidence-based practices (i.e., Classroom Observation Tool [COT]; Crawford et al., 2013) and a linked Short Term Goal Report (STGR) form to document goals/action items within each coaching session. After each coaching session, the coach emailed these forms to the teacher with links to additional resources to help the teacher meet their goals (e.g., an exemplar modeling video, active from CIRCLE Activity Manual). Screenshots of eTEEM components are in Online Appendix Figures A1–A4. The training and supervision of the classroom coach who supported all cohorts of teachers is detailed in Online Appendix Table A5 and indicates average quality of coaching delivery was adequate $M = 3.32$ ($SD = 0.52$) on a 5-point rating.

In comparison to the original Landry et al. (2017) classroom intervention, the main difference was delivery of the coaching remotely rather than in the classroom. Remote coaching required less total time (3 hour per month) than face-to-face (4 hour) because coach observations were shorter segments of instruction. Most of the eCIRCLE online courses were the same ones used in the original study, but were delivered remotely rather than with in-person facilitation of the course. Also 3 courses were added (i.e., English learners, math learning, and social-emotional supports). The CPM

progress monitoring tool contained the same items and similar on-line platform as the original study. The kits and CIRCLE Activity Manual were the same materials provided in 2017 study, but the supplemental read-aloud materials and math manipulatives were added as part of enhancements to the broader TEEM PD model implemented across a large state (Crawford et al., 2017).

2.4. Parent intervention procedures

Parents/primary caregivers randomly assigned to the ePALS treatment condition participated in about 20 hour of intervention activities with most beginning in November/December of their child's pre-k year and ending in April/May; however, there was flexibility to accommodate parent schedules (e.g., start January/February, end June/July). Given variability in caregivers' availability, parents were encouraged to complete at least 6 coaching sessions. Each family in the ePALS conditions was assigned a Family Coach who supported the primary caregiver (93% mothers; 5% fathers; 2% other such as grandmother). The first session included a home visit in which the coach: (a) explained the ePALS program and rationale, (b) loaned the family a tablet device (iPad mini with childproof case and stand), (c) supported the parent in using the ePALS mobile application (app) and videoconference software (Zoom); and (d) completed an interview to learn about the caregiver's goals for the program. At this first visit, the coach also tested the home internet signal strength (Speedtest app) and recorded a 5 minute demo video to attempt to upload it to the ePALS app. If it did not upload within 10 minute, then research staff worked with a cable Internet provider (e.g., AT&T, Comcast) to install home internet service for the duration of the study. There were 7.9% of families ($n = 50$; 2014 = 12 families; 2015 = 28 families; 2016 = 10 families) who required Internet to be paid by the grant; all others already had sufficient internet speed to access the online program.

Thereafter, weekly ePALS sessions alternated between 15 didactic online sessions (60 minute each) and 15 coaching sessions (about 30 minute each). The didactic modules introduced new topics and parent strategies with video exemplars of parents demonstrating the behaviors. The ePALS app was developed by Oregon Research Institute. The ePALS session schedule is detailed in Online Supplementary Appendix Table A6 and app screenshots are in Figure A5–A8. ePALS sessions address *responsiveness*, (attending to young children's communicative signals; responding contingently; following the child's lead in play and maintaining rather than redirecting the child's attention), *guided learning* (using scaffolding strategies to support children's language/literacy development; and using positive behavior management strategies) and creating predictable *daily routines* (e.g. mealtime, dressing, bedtime, shared book reading, toy play). The proportion of course time across topics was 38% responsiveness, 31% social-behavioral, 23% language, and 8% literacy. During the week, the parent used the tablet's camera and stand to record 2 videos (2- to 10-minutes each) of parent-child play or daily activities in which the parent practiced applying the focal strategies.

Coaches reviewed these uploaded parent-child videos and then scheduled a coaching call. During coaching videoconference the coach replayed clips of the videotaped practice to help the parent self-reflect and to discuss parents' progress toward goals. Coaching calls took place on weekdays or weekends between the hours of 7am–9pm at a time the parent selected (see Figure A8: Coach Call). Although ePALS focuses on the primary caregiver, the child's other caregiver (e.g., father) was invited to review the online content and attend midpoint and ending sessions that review and shares the focal strategies. ePALS parents also: (a) recorded reflections/goals in a notebook/manual, (b) earned online badges as they completed each course, and (c) received magnets to display in their home

as reminders summarizing key ePALS strategies. The ePALS curriculum is available in Spanish and English and was delivered in the parent's preferred language by a coach fluent in that language (33.1% selected Spanish). The training and supervision of the 19 total ePALS coaches is detailed in Online Appendix A7-A8 indicates average quality of coaching delivery was high $M = 1.87$ ($SD = 0.22$) on a 2-point rating.

In comparison to the original Landry et al. (2017) study, the main difference was delivery of the learning content and coaching remotely rather in home visits. The original study included 12 learning sessions, 6 coaching sessions, and 2 reviews with alternate caregiver, requiring approximately the same hours [20 total]; however, in the original study the parent program started earlier, in about October of each school year. A smaller proportion of families in this study requested Spanish ePALS (33%) compared to the original study (45%).

2.5. Description of counterfactual conditions

In the No eTEEM/control condition, most teachers (98%) reported using a published general curriculum for language arts/general instruction (e.g., Frog Street and Creative Curriculum); only 24% of teachers reported using a math-specific curriculum. All teachers completed items from the Early Childhood Longitudinal Study (ECLS; U.S. Department of Education, 2005) teacher questionnaire and $\geq 80\%$ reported ample instructional resources for language/literacy (e.g., books, alphabet activities) and math instruction (e.g., math manipulatives, counting or shape books). The ECLS survey included a 4-point scale asking teachers how often they taught different skills (1—"not at all", 4—"a lot"). Teachers in these Head Start classrooms reported slightly less coverage of mathematics instruction compared to language/literacy and social-emotional skills instruction. In the No ePALS/control condition, these parents (as well as ePALS parents) were given 2 children's books and a puzzle to support school readiness.

2.6. Fidelity of implementation

2.6.1. Teacher fidelity

We documented teachers' exposure to the eTEEM PD and responsiveness to coaching; detailed fidelity data are provided in Online Appendix Table A9. Exposure or dosage of PD included 2 variables - the number of online courses completed and the number of coaching cycles completed. On average, teachers completed 77.4% of online courses and participated in 78.4% of the offered coaching cycles. Within these cycles they set an average of 97.75 goals using the COT ($SD = 22.33$). Note goals are often closely linked and can be achieved in a single classroom activity (e.g., counting sets [goal 1], using manipulatives [goal 2], during small group lessons [goal 3]). Teachers and coaches recorded meeting an average 79.0% of these COT goals. The coach used a 5-point Likert scale (5-High, 1-Low) to rate each teachers' levels of engagement across these criteria: engagement during the call ($M = 4.06$, $SD = 0.92$), coachability or openness to feedback ($M = 3.91$, $SD = 1.04$), mastery of the teaching activity objectives ($M = 3.41$, $SD = 0.69$), mastery of the online course content ($M = 3.04$, $SD = 0.89$), and reflectiveness in oral responses ($M = 3.42$, $SD = 0.83$). These dosage indicators suggest adequate implementation (Hill & Erickson, 2019). In comparison to the original study, teachers completed a similar amount of coursework (79% course completion in face-to-face versus 77% in present study). However, teachers completed substantially less remote coaching cycles (78%) than in the face-to-face original study that had higher coaching completion (92%).

2.6.2. Parent fidelity

In terms of dosage, parents participated in about 6 of the 15 ePALS videoconference coaching sessions ($M = 5.90$, $SD = 4.48$, Range = 1–14; 39%). Many of these coaching sessions were moved to phone calls due to technical issues. In total (videoconference or phone) parents participated in about 44% of the offered coaching sessions. Parents completed about 57% of the online learning sessions in the ePALS app. Most parents (74%) completed the first 4 PALS learning sessions. About 62% of families continued through the sessions 5–9. However, only about half of parents completed all the remaining PALS course sessions (Sessions 10–15). On average, parents spent about an hour completing each course ($M = 65.87$ minute, $SD = 29.76$ minute, Range = 11 to 212 minute). To understand parents' responsiveness, coaches used a Likert scale to rate parent quality of engagement, knowledge acquisition and mastery of ePALS skills (max score 17). On average, these data showed adequate quality and engagement ($M = 12.24$, $SD = 2.86$). Coaches also evaluated parents adherence to the ePALS strategies demonstrated in coaching videos with a 4-point Likert rating of 10 key skills. Most parents demonstrated adequate adherence to using taught skills ($M = 31.77$, $SD = 5.48$, Range 29–39). These ePALS implementation data are detailed in Online Appendix A10-A12.

3. Results

3.1. Pre-Analysis data inspection & analytic strategy

Prior to analyses, we examined missing data patterns, possible outliers, and the distribution of all outcome variables. Several variables demonstrated non-normality (e.g., self-regulation measures not observed or reported by parents/teachers). For variables with non-normality, we used different model assumptions as appropriate (e.g., negative binomial, logistic and binary). We used maximum likelihood estimation for all analyses to account for missing data. For continuous outcomes, the group differences of least square means (LSMs) were reported and Hedges' g effect size was calculated as a measure of effect size.¹ For binary outcomes, the odds ratio was reported and used as a measure of effect size.

We conducted 3 sets of analyses to address the research questions. First, we examined the effect of the eTEEM program on teacher behaviors assessed at pretest and posttest. Therefore, the analysis of covariance (ANCOVA) model was used to assess posttest differences, controlling for levels at pretest, educational level, and years of teaching experience. Second, we assessed the effect of the ePALS program on parent behaviors, taking into account the impact of the eTEEM program, although we did not expect any influence of the eTEEM intervention on parent outcomes.² Mixed modeling was employed to control for the non-independence in data; parent and child outcomes that were nested within classrooms. An ANCOVA mixed model was used to assess posttest differences, controlling for the impact of the eTEEM program as well as pretest levels and covariates: child gender, age of children, and home language. Third, we tested the impacts of eTEEM and ePALS programs on children's outcomes. Specifically, the main effects of eTEEM and ePALS and the interaction of eTEEM and ePALS were estimated simultaneously, controlling for the pretest levels and child gender, age, and home language use.

¹ We use Hedge's g as this is what is recommended currently by the What Works Clearinghouse. We note that Hedge's g is very similar to Cohen's d in larger samples. In the current study, effect sizes were calculated using both methods; differences were 0.03 or less.

² We also re-ran analyses considering only the effect of PALS. The pattern of results were similar to the analyses that also include the main effect of eTEEM and interaction term, thus we report the full model below as this model is consistent with the models presented in Landry et al., 2017

Table 2

Teacher outcomes: descriptive statistics and results of ANCOVA.

	Pretest		Baseline difference		Posttest		Intervention effect	
	eTEEM M (SD)	Control M (SD)	LSM Diff.	Effect size (g) ^a	eTEEM M (SD)	Control M (SD)	LSM Diff.	Effect size (g) ^b
TBRS Total	2.27 (0.36)	2.20 (0.38)	0.06	0.16	2.48 (0.40)	2.18 (0.31)	0.26**	0.81
Sensitivity	2.74 (0.45)	2.55 (0.57)	0.18	0.32	2.98 (0.38)	2.58 (0.61)	0.29*	0.66
Classroom Community	2.65 (0.51)	2.50 (0.45)	0.11	0.21	2.60 (0.44)	2.38 (0.52)	0.16	0.35
Oral Language	2.54 (0.56)	2.37 (0.53)	0.18	0.31	2.88 (0.44)	2.41 (0.54)	0.43**	0.84
Book Reading Behaviors	2.25 (0.46)	2.24 (0.49)	0.02	0.03	2.58 (0.42)	2.14 (0.48)	0.42**	0.93
Phonological Awareness	1.24 (0.50)	1.30 (0.44)	−0.05	−0.10	1.54 (0.58)	1.41 (0.48)	0.24	0.47
Print and Letter Knowledge	2.02 (0.56)	2.13 (0.44)	−0.14	−0.25	2.49 (0.69)	2.15 (0.45)	0.40**	0.68
Written Expression	1.85 (0.66)	1.71 (0.42)	0.13	0.23	2.02 (0.70)	1.71 (0.47)	0.21	0.37
Math	2.45 (0.66)	2.41 (0.57)	0.06	0.08	2.59 (0.61)	2.47 (0.45)	0.04	0.07
Team Teaching	2.26 (0.64)	2.04 (0.73)	0.24	0.35	2.24 (0.91)	2.00 (0.59)	0.17	0.21
Centers	2.22 (0.78)	2.17 (0.67)	−0.01	−0.01	2.49 (0.85)	1.90 (0.57)	0.54**	0.77
Lesson Plans	2.55 (0.86)	2.56 (0.85)	0.00	0.00	2.60 (1.06)	2.53 (0.70)	0.07	0.08
CEC Total	44.52 (7.03)	43.22 (5.12)	1.40	0.21	46.87 (6.25)	42.78 (6.84)	2.74*	0.49
Center/Independent Work Areas	13.32 (2.55)	12.59 (1.91)	0.81	0.34	13.81 (2.88)	12.07 (2.57)	1.27*	0.47
Instruct. Planning & Delivery Areas	7.68 (1.72)	7.44 (1.50)	0.32	0.18	8.06 (1.61)	8.15 (1.75)	−0.24	−0.14
Literacy Areas and Descript.	9.94 (1.84)	10.11 (1.50)	−0.16	−0.09	10.65 (1.98)	9.81 (1.80)	0.86*	0.47
Meaningful Print Area & Descript.	13.58 (2.36)	13.07 (2.60)	0.43	0.16	14.35 (2.04)	12.74 (2.99)	1.04*	0.49
ELLCO Total	59.42 (10.70)	58.70 (11.19)	0.40	0.03	67.10 (10.93)	56.15 (10.09)	10.53**	1.14
Books and Book Reading	16.65 (3.62)	16.00 (3.43)	0.63	0.17	18.61 (3.22)	15.48 (2.90)	3.04**	0.98
Classroom Structure	14.48 (2.63)	14.44 (2.44)	−0.01	0.00	15.84 (2.45)	13.04 (2.28)	2.79**	1.34
Curriculum	9.06 (2.26)	8.70 (2.48)	0.29	0.11	10.10 (2.21)	8.96 (2.23)	0.92	0.45
Language Environment	11.32 (2.39)	11.04 (3.18)	0.05	0.02	13.26 (2.61)	10.56 (2.89)	2.61**	1.05
Print and Early Writing	7.90 (2.51)	8.52 (2.03)	−0.55	−0.22	9.29 (2.47)	8.11 (1.80)	1.19*	0.52
Teacher Organization								
CHAOS	36.64 (8.15)	34.73 (6.07)	−1.91	−0.34	35.55 (7.79)	31.40 (6.18)	−3.21	−0.51

Note. TBRS, Teacher Behavior Rating Scale; CEC, Classroom Environment Checklist; ELLCO, Early Language and Literacy Classroom Observation. LSM = least squares mean.

^a Effect sizes are Hedges' *g* computed using pretest raw score means and pooled standard deviations (SDs).

^b Effect sizes are Hedges' *g* computed using posttest LSMs and pooled SDs.

* $P < 0.05$.

** $P < 0.01$.

3.2. RQ1: what were the impacts of eTEEM and ePALS on adult behaviors?

3.2.1. Teacher outcomes

Table 2 provides the descriptive statistics (means and standard deviations) and ANCOVA results, including the group differences of LSMs and effect sizes (*g*), for analyses of eTEEM effects on the TBRS, CEC, and ELLCO scores. At pretest, results in Table 2 showed that there were no statistically significant differences between eTEEM and No eTEEM groups in their TBRS total score (group difference of LSMs = 0.06; $g = 0.16$), in their CEC total score (group difference of LSMs = 1.40; $g = 0.21$), or ELLCO total score (group difference of LSMs = 0.40; $g = 0.03$) as well as no pretest differences on subscale scores.

Main effects at posttest demonstrated that teachers who received eTEEM had significantly higher TBRS total score than teachers who did not receive eTEEM (group difference of LSMs = 0.26; $g = 0.81$). Regarding the TBRS subscales, we found teachers who received eTEEM had statistically significant higher scores on 5 of 11 subscales, including: Sensitivity (group difference of LSMs = 0.29; $g = 0.66$), Oral Language (LSMs = 0.43; $g = 0.84$), Book Reading Behaviors (LSMs = 0.42; $g = 0.93$), Print and Letter Knowledge (LSMs = 0.40; $g = 0.68$), and Centers (group difference of LSMs = 0.54; $g = 0.77$).

At posttest, there were statistically significant main effects for teachers who received eTEEM who displayed a greater CEC total score compared to teachers who did not receive eTEEM (group difference of LSMs = 2.74; $g = 0.49$). Group differences were significant for 3 of the 4 subscales in which eTEEM teachers had higher scores for: Center/Independent Workstation Areas (group difference of LSMs = 1.27; $g = 0.47$), Literacy Areas and Description (group difference of LSMs = 0.86; $g = 0.47$), and Meaningful Print Area (LSMs = 1.04; $g = 0.49$).

Significant main effects favoring eTEEM teachers were observed for the ELLCO total score (group difference of LSMs = 10.53; $g = 1.14$). This pattern emerged for 5 of 6 ELLCO subscales: Books and Book Reading (group difference of LSMs = 3.04; $g = 0.98$), Classroom Structure (LSMs = 2.79; $g = 1.34$), Language Environment (LSMs = 2.61; $g = 1.05$), and Print and Early Writing (LSMs = 1.19; $g = 0.52$). Finally, no significant group differences were observed for the Classroom CHAOS scale.

3.2.2. Parent outcomes

Online supplementary Table A10 provides pretest and posttest descriptives for all parent measures, while Table 3 in text provides analyses of ePALS effects on the parenting behaviors during book reading, free play sessions, and a home scale. At pretest, there were no statistically significant main effects of ePALS (g ranging from −0.22 to 0.05). In order to examine treatment specificity, the impacts of ePALS were examined considering the impacts of the eTEEM program. As expected, there were no statistically significant effects for eTEEM or the interaction except for a main effect on one of the parenting behaviors during free play, namely Contingent Responsiveness (group difference of LSMs = 0.24; $g = 0.21$).

In the shared book reading task, there were statistically significant main effects of ePALS, after controlling for pretest levels and covariates. Parents who received ePALS had a higher caregiver book reading total posttest score than parents who did not receive ePALS (group difference of LSMs = 2.71; $g = 0.79$). Specifically, ePALS parents had statistically significant higher scores on the Language Building Strategies (LSMs = 0.62; $g = 0.54$), Book Comprehension Techniques (LSMs = 0.68; $g = 0.57$), Enthusiasm and Engagement (LSMs = 0.74; $g = 0.62$), and Flexibility and Responsiveness (LSMs = 0.73; $g = 0.63$) subscales than parents in the No ePALS group. There was no statistically significant group difference for a subscale of Intrusiveness and Tone; however, there was little variation in means.

Table 3
Parent outcomes: effects of PALS on parenting behaviors.

	Intervention effect (b)			Effect size (g) ^a		
	Main Effect of ePALS	Main Effect of eTEEM	eTEEM and ePALS Interaction	Main Effect of ePALS	Main Effect of eTEEM	eTEEM and ePALS Interaction
<i>Book Reading Scales</i>						
Caregiver Book Reading Total	2.71**	0.10	−0.21	0.79	0.03	−0.06
Language Building Strategies	0.62**	−0.15	0.02	0.54	−0.13	0.02
Book Comprehension Techniques	0.68**	−0.17	−0.01	0.57	−0.14	−0.01
Enthusiasm and Engagement	0.74**	0.21	−0.26	0.62	0.18	−0.02
Contingent Responsiveness	0.73**	0.19	−0.10	0.63	0.16	−0.09
Intrusiveness and Tone	−0.04	0.00	0.13	−0.13	0.01	0.37
<i>Free Play Scales</i>						
Contingent Responsiveness	0.79**	0.24*	−0.20	0.70	0.21	−0.17
Negativity ^b	N/A	N/A	N/A	N/A	N/A	N/A
Verbal Content, Label Frequency	0.54**	0.01	−0.13	0.59	0.01	−0.14
Verbal Content, Label Quality	0.33**	0.07	0.11	0.37	0.08	0.13
Verbal Scaffold	0.75**	−0.09	0.07	0.62	−0.07	0.06
Warmth	0.78**	0.07	−0.04	0.70	0.06	−0.03
<i>Parent Organization</i>						
CHAOS	−0.02	−0.01	−0.11	−0.02	−0.01	−0.09

Note. Main effects are adjusted least square mean differences.

^a Effect sizes are Hedges' *g* computed using posttest raw score means and pooled standard deviations (SDs).

^b Negativity is a reverse scored measure. Note it has a SD close to 0 (trivial variation) and thus cannot be analyzed. CHAOS = Confusion, Hubbub and Order Scale.

* $P < 0.05$.

** $P < 0.01$.

During the free play task, there were significant main effects of ePALS on the Contingent Responsiveness (group difference of LSMs = 0.79; $g = 0.70$), Verbal Content-Label Frequency (LSMs = 0.54; $g = 0.59$), Verbal Content-Label Quality (LSMs = 0.33; $g = 0.37$), Verbal Scaffold (LSMs = 0.75; $g = 0.62$), and Warmth (LSMs = 0.78; $g = 0.70$). As shown in Table 3, the scale of Negativity had close-to-zero score variation and thus was not applicable for the analysis. The means of Negativity of 4 treatment groups were quite similar to each other at pretest (ranging from 2.93 to 2.99) and at posttest (ranging from 2.93 to 2.99), implying the group differences were unlikely to be detected at both pretest and posttest. Counter to expectations, there were no main effects of ePALS for the CHAOS scale at pretest ($g = -0.05$) and posttest ($g = -0.03$).

3.3. RQ2: did the individual and combined interventions increase children outcomes?

Online Supplementary Table A13 provides the descriptive statistics for all child outcomes and Table 4 provides the results for analyses of ePALS and eTEEM effects on the children's outcomes behaviors during book reading and free play sessions, academic outcomes as well as their social-behavioral measures. At pretest, there were statistically significant effects for the ePALS by eTEEM interaction on the Book Reading Engagement scale ($g = -0.41$) and the pretest main effect of ePALS on children's Dragon task (odds ratio = 0.54). No other pretest effects were significant (g ranging from -0.39 to 0.16). Pretest scores were controlled for in all analyses.

As mentioned, preliminary evaluation indicated that some measures had non-normal distributions, particularly the direct self-regulation measures. The Gift Delay and Waiting for Bow task pretest and posttest distributions included 60–70% of children either not waiting (0 or 1 second latency), or waiting the full time. We modeled these variables as binary distributions denoting “pass” as waiting the entire time period and all other times as failure to wait. Similarly, the DCCS and Bear/Dragon tasks pretest and posttest histograms also indicated binary distribution patterns (e.g., over 75% of children scoring either no points or all points); this is common of these self-regulation tasks in the literature (e.g., Carlson, 2005). We also dichotomized these variables denoting

“pass” as the child always correctly responding, and all other response patterns as a failure to regulate.³

3.3.1. Shared book reading behaviors

There were statistically significant main effects of ePALS, after controlling for pretest levels and covariates. No main effects of eTEEM and interaction effects were statistically significant (see Table 4). Specifically, children whose parents received ePALS had a higher child rating total score than children whose parents did not receive ePALS (group difference of LSMs = 1.18; $g = 0.57$). Regarding the book reading subscales, children in ePALS condition had statistically significant higher scores on the Book Reading Engagement (group difference of LSMs = 0.58; $g = 0.51$) and Language Use (group difference of LSMs = 0.60; $g = 0.50$) compared to those in No ePALS condition. For Shared Enjoyment, the main effect of ePALS was statistically significant, after controlling for pretest levels and covariates. That is, children in ePALS condition had a statistically significant higher level of enjoyment than those in No ePALS condition (group difference of LSMs = 0.75; $g = 0.66$). No main effects of eTEEM and interaction effects were statistically significant.

3.3.2. Behaviors during free play

At posttest, there was one significant main effect of ePALS on the scores of Goal Directed Behavior, while remaining main effects or interaction effects were not statistically significant (see Table 4). Particularly, children whose parents received ePALS had statistically significant higher scores on the Goal Directed Behavior (group difference of LSMs = 0.25; $g = 0.25$) compared to children whose parents did not receive ePALS.

3.3.3. Early academic skills

For the early literacy, language, and math measures, there were no significant main effects for eTEEM and ePALS or interaction effect of eTEEM*ePALS, see Table 4. We note trend level significance ($P < 0.10$) for: EOWPVT, and TEMA math scores. Specifically, for

³ Given that each measure displayed ceiling effects, we also analyzed the data by transforming the distribution (e.g., reverse coding time or score) so that a negative binomial model could be fit. The result pattern was similar to the reported binary models. We present results from the binary models as these models were a more accurate description of the data.

Table 4

Child outcomes: effects of TEEM and PALS on children's observed behaviors and measures.

	Intervention effect (b)			Effect size (g) ^b		
	Main Effect of ePALS	Main Effect of eTEEM	Interaction of eTEEM and ePALS	Main Effect of ePALS	Main Effect of eTEEM	Interaction of eTEEM and ePALS
1. Observed Behaviors						
<i>Book Reading Scales</i>						
Child Rating Total	1.18**	0.01	0.09	0.57	0.00	0.05
Book Reading Engagement	0.58**	0.08	−0.02	0.51	0.07	−0.02
Language Use	0.60**	−0.09	0.09	0.50	−0.08	0.08
Shared Enjoyment	0.75**	0.19	−0.11	0.66	0.17	−0.09
<i>Free Play Scales</i>						
Language Use	0.12	−0.01	0.07	0.10	−0.01	0.06
Goal Directed Behavior	0.25**	−0.05	−0.01	0.25	−0.04	−0.01
Responsiveness	0.14	−0.09	−0.09	0.11	−0.06	−0.07
Social Initiative and Communication	0.07	−0.01	−0.09	0.06	−0.01	−0.07
2. Academic Outcomes						
TOPEL Phonological Awareness	−0.15	1.13	−1.58	−0.01	0.08	−0.13
TOPEL Print Knowledge	−1.68	−0.43	−0.61	−0.15	−0.04	−0.05
EOWPVT	0.14	−0.27	2.62 ^t	0.02	−0.03	0.30
PLS Auditory/Receptive	0.11	0.50	−0.53	0.02	0.09	−0.10
PLS Expressive Communication	−0.13	−0.88	0.26	−0.02	−0.13	0.04
TEMA total	−1.03	2.23 ^t	−2.54	−0.09	0.18	−0.23
3. Self-Regulation Outcomes^a						
Gift Wrapping	−0.05	0.10 ^t	−0.07	0.80 ^a	1.68 ^t	0.68
Waiting for Bow	0.09	0.05	−0.13	1.24	1.43	0.59
DCCS	−0.07	0.08	−0.15	0.76	1.39	0.55
Bear/Dragon	0.10	0.21**	−0.34*	1.65	2.81	0.21
4. Social-Emotional Functioning						
Emotion Understanding Expressive	0.00	0.00	0.05	0.00	0.00	0.24
Emotion Understanding Receptive	0.03	0.01	−0.06	0.05	0.13	−0.27
Emotion Understanding Situational	−0.01	0.03	−0.08	−0.04	0.10	−0.27
Teacher rated CBQ Attention Focusing	0.07	0.07	0.12	0.08	0.07	0.14
Teacher rated CBQ Inhibitory Control	0.04	0.10	−0.01	0.04	0.08	−0.01
Teacher rated CBQ Impulsivity	−0.01	0.02	0.02	−0.01	0.02	0.03
Teacher rated SCBE Anxiety/Withdrawn	0.05	0.11	0.09	0.07	0.11	0.12
Teacher rated SCBE Social Competence	0.01	0.10	−0.06	0.01	0.08	−0.06
Teacher rated SCBE Anger/Aggression	0.00	0.09	−0.04	0.00	0.08	−0.05
Teacher rated SLAQ school liking	−0.05	−0.02	0.03	−0.08	−0.03	0.04
Teacher rated SLAQ school avoidance	0.05	0.08	0.03	0.08	0.10	0.04
Parent rated SCBE Anxiety/Withdrawn	0.01	0.00	0.06	0.03	0.00	0.13
Parent rated SCBE Social Competence	0.10	−0.04	0.02	0.17	−0.07	0.03
Parent rated SCBE Anger/Aggression	−0.03	−0.05	−0.05	−0.07	−0.10	−0.11
Parent rated CBQ Attention Focusing	0.12*	0.04	−0.17	0.23	0.08	−0.32
Parent rated CBQ Inhibitory Control	0.11 ^t	0.16**	−0.06	0.19	0.27	−0.10
Parent rated CBQ Impulsivity	−0.13**	0.00	−0.13	−0.25	0.00	−0.25

Note. Main effects are adjusted least square mean differences.

^a Self-regulation measures demonstrated floor and ceiling effects that were best represented by a binary model and thus those results are presented including odds-ratios to represent effect sizes.^b Effect sizes are Hedges' *g* computed using posttest raw score means and pooled standard deviations (SDs). We also ran logistic models that only accounted for the ceiling effects but the pattern of results were similar so the binary are presented. EOWPVT; Expressive One-Word Picture Vocabulary Test; PLS, Preschool Language Scale-fourth edition Auditory Comprehension and Expressive Communication scales; TOPEL, Test of Preschool Early Literacy; DCCS, Dimension Change Card Sort; TEMA, Test of Early Math Achievement; CBQ, Child Behavior Questionnaire; SCBE, Social Competence and Behavior Evaluation; SLAQ, School Liking and Avoidance Questionnaire.

* <0.05.

** <0.01 *t* <0.10.

EOWPVT scores, the eTEEM*ePALS interaction was at trend (group difference LSM=2.62, *g*=0.30, *P*=0.09), while for TEMA, there was a trend effect for eTEEM (LSM=2.23, *g*=0.18, *P* < 0.06). Although effect sizes were small, these results offer promise regarding children's early academic skills.

3.3.4. Self-regulation and social-behavioral functioning

Results for self-regulation skills are presented in Table 4. We found statistically significant effects on the score of Bear/Dragon task. Specifically, at posttest, results indicated there was a significant main effect of eTEEM (odds ratio=2.81, *P* < 0.01) and a significant ePALS by eTEEM interaction effect (odds ratio=0.21, *P* < 0.05) on the Bear/Dragon task, after controlling for pretest levels and covariates. The significant main effect of eTEEM indicated that, in general, children in the eTEEM classrooms showed greater levels of self-control compared to children in No eTEEM

classrooms. The significant interaction effect further suggested that the effect of eTEEM could be dependent on ePALS (see Table A10 for the scores of Bear/Dragon). That is, the positive effect of eTEEM was slightly stronger on children whose parents did not receive ePALS compared to children whose parents received ePALS. Note that although the interaction effect was statistically significant, its effect size was small and should be interpreted conservatively.

At posttest, results indicated there were statistically significant main effects of ePALS on the parent rated scores of Attention/Focus, Inhibitory Control, and Impulsivity, as well as one significant main effect of eTEEM on Inhibitory Control, after controlling for pretest levels and covariates. No interaction effects were statistically significant. Specifically, children whose parents received ePALS had statistically significant higher scores on the Attention/Focus (group difference of LSMs=0.12; *g*=0.23), Inhibitory

Control (LSMs = 0.12; $g = 0.21$), and a lower score on the Impulsivity (LSMs = -0.13; $g = -0.26$) compared to children whose parents did not received ePALS. For children's Inhibitory Control scores, there was also a statistically significant main effect of eTEEM (LSMs = 0.15; $g = 0.26$). That is, children in eTEEM classrooms had higher scores of Inhibitory Control compared to those who were in No eTEEM classrooms. No other self-regulation results were significant. We note trend significance ($P < 0.10$) for the Gift Wrap eTEEM main effect (odds-ratio = 1.68, $P < 0.10$). Contrary to expectations, there were no significant differences for any of the social-behavioral parent and teacher reported behaviors (g ranging from -0.27 to 0.27). This is similar to the findings in Landry et al., 2017.

4. Discussion

This replication study examined the combination of 2 evidence-based aligned classroom and home interventions in Head Start centers. Like the original study, the present study used a 4-group factorial design to determine if remote delivery of The Early Education Model (eTEEM) in classrooms and remote Play and Learning Strategies (ePALS) in the home would result in similar findings to those reported for the in-person study (Landry et al., 2017). As in the in-person study, remotely training both teachers and parents to use social-behavioral and academically responsive behaviors resulted in significant increases in the targeted adult behaviors. Also, as in the original study, each of the remote interventions resulted in positive increases in child skills, particularly social behavioral and self-regulation.

4.1. Effect of the remote interventions on parent and teacher behaviors

The theory of change for this replication study seeks to improve focal parent and teacher behaviors in order to see increases in children's school readiness skills. The focus of the home and classroom aligned interventions was on 3 key caregiver behaviors, a responsive approach, guided learning, and organization that supported predictable routines and practices. Each of these key behaviors are reported to support young children's development and together to provide an integrated approach that facilitates a broad range of academic and social-behavioral skills (Grusec & Davidov, 2010). Coordinating support for caregivers across home and school to learn to use these key behaviors was expected to enhance the impact of the interventions on children's learning as they would experience consistency and greater intensity of support in the 2 environments where they spend the greatest amount of time.

As hypothesized, this replication study found increases in the 3 key targeted adult behaviors with moderate to large effect sizes. First, we found that sensitive responsiveness showed large increases for both parents ($g = 0.63$ to 0.70) and teachers ($g = 0.66$). These adult behaviors have been found in our previous studies (Landry et al., 2008, 2017) and other work (e.g., Domitrovich et al., 2007; Guttentag et al., 2014) to support the development of children's social-behavioral and academic skills. Second, we found increases in guided learning for rich language input, verbal scaffolding, and book reading practices for teachers ($g = 0.84$ to 1.05) and parents ($g = 0.37$ to 0.62). Past research shows these adult behaviors facilitate increases in child language, literacy, attention, and self-regulation skills (e.g., Crawford et al., 2020; Lonigan et al., 2015; Mashburn, Justice, McGinty & Slocum, 2016). Third, the classroom intervention eTEEM also resulted in improvements in organization and implementation of routines with large increases in teachers establishing a classroom environment that supported learning (e.g., TBRS Learning Centers, $g = 0.77$; ELLCO Classroom Structure, $g = 1.34$). These adult outcomes generally demonstrate that remote approaches implemented in homes and classrooms

can be effective in supporting positive change in both social-behavioral and academically responsive adult practices with pre-k children.

However, one of the 3 key adult behaviors - organization and predictable routines - increased significantly in eTEEM, but not in ePALS as measured with the home CHAOS scale. This may be explained by the different intensity of emphasis placed on this key behavior in the classroom versus the home. Classroom coaches worked remotely with teachers to organize their classrooms to support children to use materials independently, establish a schedule and routines, and develop strategies for helping children understand and adhere to the routines and continued to support the use of these strategies throughout the year. In contrast, ePALS coaches discussed the importance of predictability in schedules and routines and implementing activities that supported their children to feel part of the family community but this key behavior did receive less focus as a weekly goal for most families. The one exception was finding time for shared book reading as a predictable family activity that did increase in quality and quantity in ePALS. Also, the CHAOS scale that measured predictability and organization in the home may not have been sensitive to the key behaviors ePALS coaches were trying to support with their families.

Strategies implemented across the home and classroom interventions that might explain increases in similar adult behaviors included providing support for adult learning through coaching and training modules that showed believable models interacting with children using the targeted behaviors. Based on research on adult learning (Bransford, Brown & Cocking, 2000), parents and teachers were actively engaged in the process of implementing change by identifying the behaviors with their coach that they thought needed to improve. For both eTEEM and ePALS, the adults selected the behaviors they wanted to practice across the next week and self-reflected on their behaviors by reviewing videos; video reflection with a supportive coach is an important component of both interventions (cf. Stover et al., 2015). In sum, ongoing training, opportunities for practice, and reflecting on their own behavior are key ingredients of both interventions that may help explain similar increases in parent and teacher behaviors.

4.2. Comparison of adult intervention & effects in original and replication study

The general magnitude of effects for adult responsiveness were large ($g = 0.66$ to 0.70) in comparison to moderate effects ($d = 0.44$ to 0.48) in the original, face-to-face study (Landry et al., 2017). Although the delivery approach differed across the original study and this remote study, there were many characteristics that remained the same. In both interventions adults were exposed to weekly sessions that presented the theory behind the 3 key behaviors accompanied by videos of parents or teachers using the targeted behaviors in interactions with young children. These sessions were used by the coaches to generate discussion about what resonated with the adult learner and what didn't fit with their beliefs and own practices. This approach, whether implemented in-person or remotely, seemed to facilitate the development of a trusting relationship between the coach and caregiver; such relational processes are characteristic of effective family-school interventions (Sheridan, Smith, Moorman Kim, Beretvas & Park, 2019). Thus, while the delivery of this replication study was remote, supportive relationships could still be established between a coach and the adult learner. High ratings by coaches on factors like parents' and teachers' level of engagement with their coach, openness to feedback, overall reflectiveness in oral responses provide some evidence of the positive nature of the teacher/coach relationship.

As in the in-person study, the remote classroom and home interventions paid a great deal of attention to documenting par-

Table 5
Comparison of key findings in the original and current study.

Component	Original study	Current study	Present replication comparison
Teacher Fidelity	79% course completion 92% coaching sessions	77% course completion 78% coaching sessions	Similar online coursework, but reduced coaching completion in remote delivery
Parent Fidelity	$M = 2.86$ engagement rating (3-point) $M = 2.54$ skill mastery (3-point) 78% combined learning + coaching sessions	$M = 2.72$ engagement rating (3-point) $M = 3.41$ skill mastery (3-point) 57% of online learning sessions 39 to 44% of coaching sessions	Similar parent engagement and learning, but lower completion of coaching in remote delivery
Teacher Attrition	11.4% attrition	10.5% attrition	Similar across studies
Parent Attrition	30% attrition	28% attrition	Similar across studies
Teacher Outcomes	$d = 0.71$ Total TBRS	$g = 0.81$ Total TBRS	Similar, slightly larger remote teacher effects
Parent Outcomes	Book Reading $d = 0.44$ responsiveness $d = 0.55$ language building strategies	Book Reading $g = 0.63$ responsiveness $g = 0.54$ language building strategies	Similar remote parent effects
Child Social-behavioral Outcomes	$d = -0.05$ TEEM self-regulation (gift delay wrap) $d = 0.19$ PALS self-regulation (gift delay wrap) $d = 0.01$ TEEM bear/dragon $d = 0.09$ PALS bear/dragon	$g = 1.68$ eTEEM self-regulation (gift delay wrap) $g = 0.80$ ePALS self-regulation (gift delay wrap) $g = 2.81$ TEEM bear/dragon $g = 1.65$ PALS bear/dragon	Larger effects for remote than face-to-face for child self-regulation and other social-behavioral outcomes
Child Academic Outcomes	$d = 0.10$ TEEM literacy (print knowledge) $d = 0.14$ PALS literacy (print knowledge) $d = 0.05$ TEEM language (composite) $d = -0.18$ PALS language (composite)	$g = -0.04$ eTEEM literacy (print knowledge) $g = -0.15$ ePALS literacy (print knowledge) $g = 0.09$ eTEEM language (PLS-Receptive) $g = 0.02$ ePALS language (PLS-Receptive)	Similar, small and mostly non-significant effects for children in both studies

ents' and teachers' understanding and use of the target behaviors including: (a) coaches use of the Classroom Observation Tool to establish goals with the teachers and measure when these goals were met, (b) parents inviting other family members into the session and with the guidance of their coach explaining and/or demonstrating the responsive strategies they were learning, (c) the use of a fidelity checklist identifying the level of learning filled out after every sessions by coaches. Table 5 summarizes key findings across studies and shows teachers completed more than 3 quarters of the coursework in both studies (79% in face-to-face; 77% in remote). However, both teachers and parents completed substantially less of the offered coaching sessions in this remote delivery with teachers receiving 78% of offered coaching sessions and parents completing about 44% of the coaching and 57% of the online curriculum. Coaches who participated in both studies anecdotally reported that this decrease was because they could more readily control face-to-face sessions in which the initial burden is on research staff to appear in homes and classrooms.

A disappointing finding was that parent attrition remained fairly similar across studies and was substantial (28% attrition in remote; 30% attrition in face-to-face). We expected that a more convenient, remote approach could facilitate parent retention through the full course of the study. Anecdotally, parents praised the flexibility of this remote approach and their time spent viewing online learning sessions ($M = 65$ minute) suggests high engagement with the content. Although we did not realize our goal of delivering the full treatment dosage to all adults, the fact that parent outcomes were of similar or larger magnitude despite reduced implementation suggests promise of the online approach.

4.3. Child intervention effects & comparisons across studies

We discuss child outcomes in order of the magnitude of effects, beginning with the areas with larger intervention impacts and then

the areas where we did not observe expected effects. The largest impacts were observed for children's self-regulation and behaviors skills ($d = 0.19$ to 0.27) and there were only non-significant trends observed for academic outcomes ($d = 0.18$ to 0.30).

4.3.1. Self-regulation and behavioral skills

There were similarities in some of the findings for children's outcomes across the 2 studies. In both studies there were more significant child findings in the behavioral and self-regulation domains of development than in academic skills. However, in general these findings were somewhat stronger for the remote interventions. For example, parents who received ePALS reported higher scores for children's attention focus ($g = 0.23$), inhibitory control ($g = 0.21$), and their impulse control ($g = -0.26$) compared to parents who did not receive ePALS and teachers in eTEEM reported better scores for children's inhibitory control ($g = 0.26$). This was in contrast to the in-person study where there were no effects on these behaviors for the classroom intervention and parents receiving PALS did not report differences on impulse control and effects on attention focus and inhibitory control were small ($d = 0.15$).

There were significant findings for self-regulation skills assessed with direct measures for both studies. However, they differed in the magnitude of the effects and the type of self-regulation skills impacted by the home versus classroom intervention. In both studies, there were increases for the type of self-regulation described by some theorists as "hot" regulation skills assessed with the Gift Wrap task (Kochanska et al., 2000). However, in the in-person study these increases were seen for children whose parents received PALS but not for those who were in the TEEM classrooms. While in the remote study, noteworthy trends with meaningful effect sizes were seen for children in eTEEM classrooms ($g = 1.68$) and for children in ePALS ($g = 0.80$). The greater magnitude of effects for responsiveness in the remote study for both teachers and parents may explain these effects, as responsiveness is often

reported as supporting children's self-regulation development as it promotes children feeling supported and secure, thus allowing them to explore their environments and extend their learning experiences.

Children who experienced the remote classroom intervention showed a significant increase in self-regulation skills ($g = 2.81$) that are often described as “cool” aspects of these skills. This finding was not seen for children in the remote parent intervention or for children in TEEM classrooms in the original study. It was unexpected that the positive eTEEM effect was slightly stronger for children who did not receive ePALS but the effect size was small. It is not clear why the children whose parents did not receive ePALS but were in eTEEM classrooms would show larger gains in this self-regulation behavior and the finding does not fit with the other positive social-behavioral findings seen for children in ePALS.

One explanation for the eTEEM effect on “cool” skills and only a trend for the “hot” regulation Gift Wrap task might be that the end of the pre-k year was more sensitive to change in “cool” regulation. Cool self-regulation skills are described as affectively neutral, reflective and later developing compared to “hot” regulation skills that are affectively engaged and are early developing (Willoughby, Kupersmidt, Voegler-Lee & Bryant, 2011). In fact, in the present study, there were more ceiling effects and few floor effects at pretest for the Gift Wrap task in contrast to Bear/Dragon that showed many pretest floor effects. Reuda et al. (2005) report that children between the ages of 4 to 6 years are more likely to show increases in “cool” regulation tasks than younger or older children. Thus, the pre-k year may be particularly sensitive to change in self-regulation skills that require the executive attention (Holmboe & Johnson, 2005).

There are only a limited number of studies that provide evidence that “cool” self-regulation skills can be enhanced with interventions, making our finding of a positive eTEEM effect interesting. The Tools of the Mind preschool curriculum that has shown positive effects on “cool” self-regulation emphasizes instructional activities that scaffold individual learner needs using dynamic assessments to guide the learning experiences. It includes play and small group learning that promote attention and working memory skills as well as reflection on learning (Diamond, Barnett, Thomas & Munro, 2007). The eTEEM program has a number of similar characteristics, including the use of dynamic assessments to guide scaffolded learning, play in learning centers and small group activities, and an emphasis on reflection on problem solving. These program components as well as the teachers' increases in responsiveness and guided learning are potentially important for understanding the eTEEM effect on the Bear/Dragon task.

4.3.2. Academic skills

Contrary to our hypotheses, there were no significant findings for the literacy or math measures. There were noteworthy, but non-significant trends for math on the TEMA for eTEEM ($P < 0.06$, $g = 0.18$) and an ePALS by eTEEM interaction ($P < 0.09$, $g = 0.30$) for the EOWPVT. The potentially meaningful trend for an interaction for expressive vocabulary provides some additional support for providing aligned interventions across the classroom and home. Parents in ePALS and teachers in eTEEM showed large increases in their language support in interactions with children as well as in shared book reading activities that together seemed to support vocabulary growth. In contrast, the lack of findings for children's literacy skills (print knowledge and phonological awareness) in eTEEM classrooms is most likely due to small or insufficient increases in teachers' instruction of this skill domain with no differences between the control and eTEEM teachers at posttest. There also were no posttest differences for the 2 groups for written expression or math. These instructional areas did not show significant posttest differences in the in-person study either, perhaps

suggesting these are difficult areas to change for Head Start teachers, rather than the result of a remote versus in-person program.

4.3.3. Synergistic effects and magnitude of effects

We looked for potential interactions that would suggest a synergistic effect of the combined interventions, but found little significant evidence of this type of greater combined effect for both interventions. There was one trend for a synergistic effect on the EOWPVT ($g = 0.30$). But results overall suggest that other approaches may be necessary to produce synergies when intervening across school and home environments; for example, perhaps approaches in which the parent and teacher form a partnership to support their child's learning (e.g., Brotman et al., 2016; Garbacz, Stormshak, McIntyre & Kosty, 2019). But the added value of such approaches is inconsistent (for review see Grindal et al., 2016).

4.4. Summary of replication procedures & findings

A chief aim of science is to accumulate knowledge in ways that inform practice. Conceptual replication studies such as this accomplish this aim by examining if similar results from a previous study can be reproduced when a study procedure is systematically varied (Coyne et al., 2016; Hedges & Schauer, 2018; Schmidt, 2009). We reproduced the pattern of effects found in the original study across several areas (see Table 5) and executed the replication procedures largely as intended, attending to the key aspects of replications (IES & NSF, 2018).

4.4.1. Population

As detailed in Online Appendix Table A1 and A2, we recruited very similar Head Start samples across studies. Two noteworthy differences were: (a) this study included only one urban locale, whereas the original study included Head Start centers at 2 urban sites within this state, and (b) the present teachers had somewhat higher levels of education.

4.4.2. Intervention components

Although the interventions maintained their broad focus on improving children's school readiness, we made some modifications to the classroom coursework sequence and topics. That is we expanded the topics from the original focus on general pedagogy, language, and literacy to include additional topics such as mathematics and social-emotional skills. The primary reason for this was new priorities of our education partners.

4.4.3. Intervention implementation

We successfully contrasted the intervention delivery mechanism in this remote study. We only required 15 technical assistance visits to homes to help parents utilize their tablet or the ePALS online sessions. Teachers required less technical support, but for both teachers' and parents' video conferencing software sometimes presented challenges if Internet speed was inadequate.

4.4.4. Outcome measures and analytic approach

We used the same outcome measures and comparable analytic procedures across studies. Additional teacher outcomes were added to support the theory of change. Based on current recommendations (What Works Clearinghouse), we computed different effect size statistics (Hedge's g versus past Cohen's d), but we were able to make comparisons across studies.

4.5. Implications for remote intervention delivery

This study was one of few to examine the remote delivery of conceptually aligned teacher and parent interventions. The findings highlight remote delivery as a promising way to deliver interventions to both teachers and parents. In other studies that have contrasted modes of delivery (i.e., remote versus face-to-face), researchers have found few to no differences on teacher- and child-level outcomes (Crawford et al., under review; Powell et al., 2010). Collectively, this suggests that remote delivery of interventions has important implications for improving the sustainability and scalability of interventions, as there are unique affordances of remote delivery that include greater convenience, engagement, and individualization for participants (Bates et al., 2019; Dede et al., 2009). Future studies should compare the cost effectiveness of the original face-to-face delivery with the current remote delivery approach (cf. Knight et al., 2019), which may be one way to determine whether remote delivery approaches are able to be sustained and scaled in classrooms and homes.

4.6. Limitations and additional future directions

Due to many efforts to keep families in the study, we did have lower parent attrition (28%) than the in-person study (30%), although not significantly less. It should be noted that these attrition rates are considerably lower than many home visiting parent programs that report attrition as high as 50% (McGuigan, Katzev & Pratt, 2003; Wagner, Cameto & Gerlach-Downie, 1996). As there continue to be technological advances, future remote intervention approaches may see fewer challenges and major improvements in ways to engage families and teachers in programs that support their responsive interactions with young children.

The lack of strong increases in the literacy and math domains is a limitation that was seen in this remote study as well as the in-person study. As eTEEM requires teachers to work on improving many literacy and math instructional approaches as well as numerous other practices including, but not limited to, being responsive to children's needs and interests, establishing predictable routines, and providing rich language input, more than one school year is probably required to see significant increases in all of these areas.

5. Conclusion

There is limited support from other aligned home and classroom studies of an added benefit of the 2 interventions provided simultaneously versus either one alone (Bierman et al., 2015; Justice, Logan & Damschroder, 2015; Lonigan & Whitehurst, 1998). This remote aligned home and classroom study also did not find strong support for an additive effect for social-behavioral or academic skills. The 2 interventions individually impacted a number of child skills with ePALS and eTEEM resulting in increases in numerous aspects of children's regulation and behavioral control; increases that may be due, in part, to the large increases in teachers' and parents' responsiveness and guided learning behaviors. The results suggest promise for programs adopting a remote approach without compromising results. However, the challenge of supporting pre-k teachers to develop highly effective and responsive practices necessary to support young children at high risk to be kindergarten ready is great. Future studies may need to provide more than one year of intervention support for teachers to integrate instructional practices at a high enough level to see strong child outcomes across social-behavioral and academic domains.

Declaration of competing interest

None.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecresq.2021.03.007](https://doi.org/10.1016/j.ecresq.2021.03.007).

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